HW #7 — Solutions

Note Title 3/9/2009

6-6. the entropy for a classical ideal gas of N atoms is

$$\sigma = N\left[\log\left(\frac{n_Q}{n}\right) + \frac{5}{2}\right]$$

For two distinguishable gases,

before mixing, $\sigma_i = N\left[\log\left(\frac{n_{QA}}{n_i}\right) + \log(n_{QB}/n_i) + 5\right]$ since $\sigma = \sigma_A + \sigma_B$

where $n_i = \frac{N}{V}$

After mixing, $n_i \to n_f = \frac{N}{2V}$

$$\Rightarrow \quad \sigma_f = N\left[\log\left(\frac{n_{QA}}{n_f}\right) + \log\left(\frac{n_{QB}}{n_f}\right) + 5\right]$$

$$= \sigma_i + 2N\log 2$$

For two indistinguishable gases.

After mixing, the two gases are just a single gas with the same concentration. so $n_f = n_i = \frac{N}{V}$. the entropy is proportional to N. so.

$$\sigma_f = 2N\left[\log(n_Q/n_i) + 5/2\right] = \sigma_i$$

6-7. a) From (3.25) The pressure in the "s" state: $P_s = -\left(\frac{\partial \epsilon_s}{\partial V}\right)_N$

$\Rightarrow$ the Average pressure is

$$P = \langle p \rangle = \sum_s P_s \, p(\epsilon_s) = -\frac{1}{Z}\sum_s \left(\frac{\partial \epsilon_s}{\partial V}\right)_N \exp(-\epsilon_s/\tau)$$

↳ the probability of finding the system in "s" state.

b) The energy of ϵ_s is the sum over energies of all occupied orbitals

$$\epsilon_s = \sum_i \frac{\hbar^2 k_i^2}{2m}, \quad \text{where } k_i = i \cdot \frac{\pi}{L}$$

$$\Rightarrow \epsilon_s \propto \frac{1}{L^2} = V^{-\frac{2}{3}}$$

$$\Rightarrow \left(\frac{\partial \epsilon_s}{\partial V}\right)_N = -\frac{2}{3}\frac{\epsilon_s}{V}$$

c) From the above eqs, $P = \frac{2}{3V}\frac{1}{Z}\sum_s \epsilon_s \exp(-\epsilon_s/\tau) = \frac{2U}{3V}$. Q.E.D.

6.9. From eq.(44) The partition function of the internal states is

$$Z_{int} = 1 + \exp(-\Delta/\tau)$$

$$n_Q^* = n_Q Z_{int} = n_Q [1 + \exp(-\Delta/\tau)]$$

n_Q is defined as before. It's the quantum contentration for the monatomic ideal gas of atoms with no spin

a) From eq(48),

$$\mu = \tau \log(n/n_Q^*)$$

b) From eq. (24) and eq(49)

$$F = F_{ext} + F_{int} = N\tau[\log(n/n_Q) - 1] - N\tau \log Z_{int}$$
$$= N\tau[\log(n/n_Q^*) - 1] \quad \text{as expected!}$$

So when take into account the internal degree of freedom, we need simply substitution $n_Q \to n_Q^*$ to get μ & F.

c) From eq. (50) & (34)

$$\sigma_{int} = \frac{\partial}{\partial \tau}(N\tau \log Z_{int}) = N \log Z_{int} + \frac{N\Delta}{\tau} \frac{1}{\exp(\Delta/\tau) - 1}$$

$$\Rightarrow \sigma = \sigma_{ext} + \sigma_{int} = \underbrace{N \log(n_Q^*/n) + \frac{\Delta/\tau}{\exp(\Delta/\tau) - 1}}_{\text{not the same as simply replacing } n_Q \text{ by } n_Q^*.} + \frac{5}{2}$$

d) F_{int} does not depend on Volume, so

$$p = -\left(\frac{\partial F}{\partial V}\right)_{\tau, N} = -\left(\frac{\partial F_{ext}}{\partial V}\right)_{\tau, N} = N\tau/V$$

e) $U_{int} = F_{int} + \tau\sigma_{int} = \dfrac{N\Delta}{\exp(\frac{\Delta}{\tau}) + 1}$, not depend on V or P.

$$\Rightarrow C_{int} = \frac{d}{d\tau} U_{int} = \frac{N\Delta^2}{2} \frac{\exp(\Delta/\tau)}{[\exp(\frac{\Delta}{\tau}) + 1]^2}$$

$$\Rightarrow C_p = \frac{3}{2}N + C_{int}.$$

6-10

a).

$$dU = TdS - pdV$$

$$\Rightarrow \quad C_V dT = TdS - pdV$$

In Isentropic process, $dS = 0$

$$\Rightarrow \quad C_V dT + pdV = 0$$

Since $PV = NT \Rightarrow dT = \frac{1}{N}(pdV + Vdp)$

1st we eliminate T, we get

$$\Rightarrow \quad \frac{C_V}{N}(pdV + Vdp) + pdV = 0$$

$$\Rightarrow \quad (C_V + N)pdV + C_V \cdot V dp = 0$$

$$\Rightarrow \quad \frac{C_V + N}{C_V}\frac{dV}{V} + \frac{dp}{p} = 0$$

$$\Rightarrow \quad \boxed{\gamma \frac{dV}{V} + \frac{dp}{p} = 0} \quad \text{(A)} \quad \text{where } \gamma = \frac{C_V + N}{C_V} \equiv \frac{C_p}{C_V}$$

We can also eliminate P, or V, we then get the corresponding Eqs:

$$\boxed{\frac{dT}{T} + (\gamma - 1)\frac{dV}{V} = 0}$$

and

$$\boxed{\frac{dp}{p} + \frac{\gamma}{1-\gamma}\frac{dT}{T} = 0}$$

Obviously, these applies even if the molecules have internal d.o.f.

b). From eq (A) $\frac{dp}{dV} = \left(\frac{\partial P}{\partial V}\right)_\sigma = -\gamma \frac{P}{V} \Rightarrow B_\sigma = \gamma P$

For an isothermal change, $P = NT/V$

$$\left(\frac{\partial P}{\partial V}\right)_T = -NT/V^2 = -P/V, \Rightarrow B_T = P.$$